YOUNES ARJDAL

Seeking a Master's Internship starting March 2027 · Machine Learning, Artificial Intelligence and Data

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PROJECTS

Structured Outputs for LLMs: JSON Schema & Constrained Decoding

M1 MIND Research Project, Sorbonne University / LIP6

2026 Pair project

- Developed an experimental pipeline to improve the reliability of structured output generation with LLMs. Integrated constrained decoding frameworks (**Outlines**, **Guidance**) with **Llama-3.1**, performed automatic validation using **JSON Schema**, and analyzed limitations related to complex schemas, timeouts, and compliance errors.

Information Retrieval Engine: BM25 & Guided Traversal

M1 MIND, Information Retrieval course

2025 Team of three

- Reimplemented a sparse search engine inspired by a SIGIR paper: built an **inverted index** from scratch, implemented **BM25** scoring, integrated neural scores, and reproduced the **Guided Traversal** strategy. Worked on core retrieval mechanisms including posting lists, top-K retrieval, query-document scoring, and efficiency-relevance trade-offs.

Road Accident Severity: Data Enrichment & Interpretable ML

M1 MIND, DALAS course

2025 Pair project

- Built an analysis pipeline to study the impact of weather conditions on road accident severity. Combined open accident data with historical weather data, automated data collection using **Selenium**, performed spatio-temporal processing with **PostgreSQL/PostGIS**, followed by preprocessing, modeling, and interpretability analysis to assess the impact of weather-related variables.

Bioacoustic Classification: Multi-label Audio ML

M1 MIND, Kaggle competition

2026 Individual project

- Designed a multi-label audio classification pipeline to identify species from Pantanal soundscapes. Extracted classical audio descriptors (**MFCC**, deltas, spectral features, RMS, chroma), trained One-vs-Rest models across **234 classes**, handled label imbalance, and analyzed the domain gap between focal recordings and test data.

Classical NLP & CamemBERT Fine-tuning

M1 MIND, RITAL course

2025 Individual project

- Compared classical NLP approaches with **CamemBERT** fine-tuning for text classification and speaker identification tasks on imbalanced datasets.

LANGUAGES

French	Native
Berber	Native
English	B2
German	B1

INTERESTS

Chess
1800 Elo

PROFILE

Second-year Master's student in MIND (Machine Learning, AI & Data) at Sorbonne University, seeking a final-year internship starting in March 2027. Through my projects, I have worked on a range of machine learning, NLP, and data-related problems, with a particular interest in experimentation and model evaluation. I have worked on end-to-end projects covering data preparation, model training and fine-tuning, benchmarking, interpretation, and error analysis on real-world datasets.

EDUCATION

M2 MIND, Machine Learning, AI & Data

Sorbonne University

2026 – 2027

M1 MIND, Machine Learning, AI & Data

Sorbonne University

2025 – 2026

International Exchange Semester

Université de Montréal (UdeM), Canada

Fall 2024

Bachelor's Degree in Computer Science

Sorbonne University

2022 – 2025

TECHNICAL SKILLS

Programming & Environment
Python, SQL, C, JavaScript, Bash, Linux, Git

Machine Learning / AI
Supervised and unsupervised learning, ensemble learning, probabilistic models, dimensionality reduction, recommender systems, fundamentals of reinforcement learning

NLP, LLMs & Retrieval
Text classification, fine-tuning, structured outputs, constrained decoding, information retrieval

Libraries & Frameworks
NumPy, Pandas, scikit-learn, PyTorch, Hugging Face Transformers, CatBoost, XGBoost, Librosa

Data Engineering
PostgreSQL/PostGIS, Selenium, web scraping, preprocessing, feature engineering, geospatial and audio data